

Buck-Boost Converter Problems

1.

The circuit shown in Figure 1 is a buck-boost converter. It is stepping from an input voltage of 12 volts to an output voltage of 15 volts.

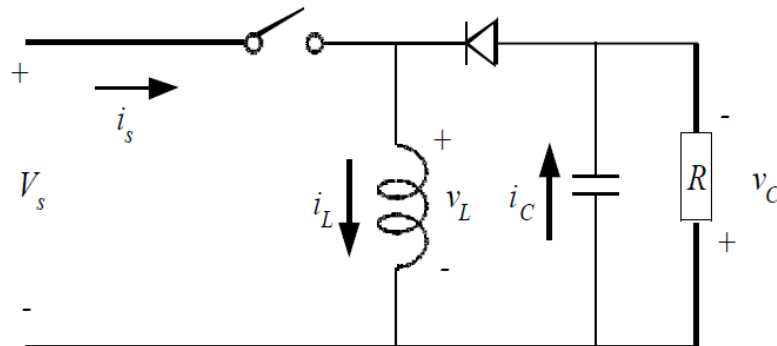


Figure 1: Buck-boost converter

- a. Assuming the inductor current is continuous by a reasonable margin, sketch waveforms for the circuit, showing the inductor voltage v_L , the inductor current i_L , the supply current i_s , the capacitor current i_C , and the expected capacitor voltage ripple. Make sure the timing is marked clearly.
 - b. Derive an expression for the output voltage of this converter as a function of the input voltage and the duty cycle D .
2. The inductor used in the buck-boost converter has finite series resistance. Draw a plot of output voltage vs duty ratio for both the ideal inductor and non-ideal inductor case.
3. What effect does finite semiconductor switch on-resistance and switching times have on converter operation and performance?